PECH GROUP HOLDINGS LTD

Technology & Infrastructure Enablers for Africa

PECH GROUP HOLDINGS LTD

AI Model Catalog — Complete Reference

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1. LICENSE POLICY

PECH only deploys models with Apache-2.0 or MIT licenses. This ensures we can: - Rebrand model outputs under the PECH brand - Modify model weights (fine-tune, quantize) - Close-source derivative products - Sell AI features as paid subscriptions - Embed models in proprietary services

Models explicitly excluded: | Model | License | Why Excluded | |-------|---------|-------------| | Llama 3 / 3.1 / 3.2 | Meta Community License | Cannot rebrand; 700M MAU limit | | Gemma 2 / 3 | Google Use Policy | Redistribution restrictions | | YOLOv8 (Ultralytics) | AGPL-3.0 | Must open-source derivative code | | Stable

Diffusion 3+ | Stability AI License | Non-commercial restrictions |

2. LARGE LANGUAGE MODELS (LLMs)

These are the primary models powering chatbot features, document generation, analysis, and AI assistants across all 10 PECH verticals.

Models Available (when multiple can do the same task, all listed)

Model	Parameters	License	VRAM (FP16)	VRAM (Q4)	Download
Qwen2.5-72B	72B	Apache-2.0	144 GB	40 GB	huggingface.co/Qwen/Qwen2.5-72B
Qwen2.5-32B	32B	Apache-2.0	64 GB	18 GB	huggingface.co/Qwen/Qwen2.5-32B
Qwen2.5-14B	14B	Apache-2.0	28 GB	8 GB	huggingface.co/Qwen/Qwen2.5-14B
Qwen2.5-7B	7B	Apache-2.0	14 GB	4.5 GB	huggingface.co/Qwen/Qwen2.5-7B
Qwen3-8B	8B	Apache-2.0	16 GB	5 GB	huggingface.co/Qwen/Qwen3-8B
Qwen3-4B	4B	Apache-2.0	8 GB	2.5 GB	huggingface.co/Qwen/Qwen3-4B
Mistral-7B-v0.3	7B	Apache-2.0	14 GB	4.5 GB	huggingface.co/mistralai/Mistral-7B-v0.3
Mixtral-8x7B	46.7B (MoE)	Apache-2.0	93 GB	26 GB	huggingface.co/mistralai/Mixtral-8x7B-v0.1
Phi-4	14B	MIT	28 GB	8 GB	huggingface.co/microsoft/phi-4
Phi-3-Medium	14B	MIT	28 GB	8 GB	huggingface.co/microsoft/Phi-3-medium-128k-instruct
DeepSeek-R1	671B (MoE)	MIT	1.3 TB	400 GB	huggingface.co/deepseek-ai/DeepSeek-R1
DeepSeek-R1-Distill-Qwen-7B	7B	MIT	14 GB	4.5 GB	huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-7B

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
General assistant (production)	Qwen2.5-14B	Best quality-to-VRAM ratio; fits single RTX 4090 in Q4; strong multilingual	Phi-4 (14B, MIT)
General assistant (starter)	Qwen2.5-7B	Runs on 5 GB VRAM quantized; good enough for Phase 1	Mistral-7B-v0.3
Complex reasoning	Qwen2.5-32B (Q4)	Strong chain-of-thought; fits in 18 GB Q4	DeepSeek-R1-Distill-7B
Maximum quality	Qwen2.5-72B (Q4)	Best open-source quality; needs cloud GPU or 2× RTX 4090	Mixtral-8x7B
Budget/edge deployment	Qwen3-4B	Newest architecture; 2.5 GB Q4; good for mobile/edge	Phi-3-Mini (3.8B)

PECH Recommendation: Start with **Qwen2.5-7B** (Phase 1 on starter hardware), upgrade to **Qwen2.5-**

14B (Phase 2), then **Qwen2.5-32B/72B** (Phase 3 with production cluster or cloud).

3. SMALL LANGUAGE MODELS (SLMs)

For edge devices, IoT gateways, mobile apps, and lightweight tasks where large LLMs are overkill.

Model	Parameters	License	VRAM (FP16)	VRAM (Q4)	Download
SmolLM2-1.7B	1.7B	Apache-2.0	3.4 GB	1 GB	huggingface.co/HuggingFaceTB/SmolLM2-1.7B
SmolLM2-360M	360M	Apache-2.0	720 MB	200 MB	huggingface.co/HuggingFaceTB/SmolLM2-360M
TinyLlama-1.1B	1.1B	Apache-2.0	2.2 GB	700 MB	huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0
Phi-3-Mini	3.8B	MIT	7.6 GB	2.3 GB	huggingface.co/microsoft/Phi-3-mini-4k-instruct
Qwen2.5-1.5B	1.5B	Apache-2.0	3 GB	1 GB	huggingface.co/Qwen/Qwen2.5-1.5B
Qwen2.5-0.5B	0.5B	Apache-2.0	1 GB	350 MB	huggingface.co/Qwen/Qwen2.5-0.5B
H2O Danube3-500M	500M	Apache-2.0	1 GB	350 MB	huggingface.co/h2oai/h2o-danube3-500m-chat

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
IoT gateway / edge	Qwen2.5-0.5B	Smallest with usable quality; 350 MB Q4	H2O Danube3-500M
Mobile app assistant	SmolLM2-1.7B	Best quality under 2B params; HuggingFace backed	Qwen2.5-1.5B
Lightweight chatbot	Phi-3-Mini (3.8B)	Punches above weight class; MIT license	TinyLlama-1.1B
Embedded/constrained	SmolLM2-360M	Runs on CPU; under 200 MB Q4	Qwen2.5-0.5B

PECH Recommendation: SmolLM2-1.7B for mobile/tablet apps, **Qwen2.5-0.5B** for IoT edge devices.

4. CODE GENERATION MODELS

For the Developer Platform, AI coding assistant, code review, and internal development tools.

Model	Parameters	License	VRAM (Q4)	Download
DeepSeek-Coder-V2-Lite	16B (MoE)	MIT	9 GB	huggingface.co/deepseek-ai/DeepSeek-Coder-V2-Lite-Instruct
Qwen2.5-Coder-7B	7B	Apache-2.0	4.5 GB	huggingface.co/Qwen/Qwen2.5-Coder-7B-Instruct
Qwen2.5-Coder-14B	14B	Apache-2.0	8 GB	huggingface.co/Qwen/Qwen2.5-Coder-14B-Instruct
StarCoder2-7B	7B	Apache-2.0	4.5 GB	huggingface.co/bigcode/starcoder2-7b
		Google		

CodeGemma-7B	7B	Terms	4.5 GB	*(excluded — Google license restrictions)*

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
AI coding assistant	Qwen2.5-Coder-14B	Best code quality in Apache-2.0 class; fits RTX 4090 Q4	DeepSeek-Coder-V2-Lite
Code completion (lightweight)	Qwen2.5-Coder-7B	Fast; 4.5 GB Q4; great for autocomplete	StarCoder2-7B
Code review / analysis	DeepSeek-Coder-V2-Lite	Strong reasoning about code; MIT license	Qwen2.5-Coder-14B

PECH Recommendation: Qwen2.5-Coder-7B for Phase 1 (autocomplete, simple tasks), upgrade to **Qwen2.5-Coder-14B** for Phase 2 (full coding assistant for developer platform).

5. SPEECH-TO-TEXT MODELS

For voice commands, customer support transcription, voice-to-edit features, and accessibility.

Model	Size	License	Languages	VRAM	Download
Whisper Large-v3	1.5B	MIT	100+ languages	4 GB	huggingface.co/openai/whisper-large-v3
Whisper Medium	769M	MIT	100+	2 GB	huggingface.co/openai/whisper-medium
Whisper Small	244M	MIT	100+	1 GB	huggingface.co/openai/whisper-small
Whisper Base	74M	MIT	100+	500 MB	huggingface.co/openai/whisper-base
whisper.cpp	Same	MIT	100+	CPU	github.com/ggerganov/whisper.cpp
Faster-Whisper	Same	MIT	100+	2 GB	github.com/SYSTRAN/faster-whisper

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Production transcription	Faster-Whisper (Large-v3)	4× faster than original; same accuracy; CTranslate2 backend	Whisper Large-v3 (original)
Real-time voice input	Faster-Whisper (Medium)	Good accuracy with low latency; 2 GB	Whisper Small
Edge / mobile	whisper.cpp (Base)	Pure CPU; C++ optimized; runs on phones	Whisper Small
Nigerian languages	Whisper Large-v3	Best multilingual coverage (Yoruba, Hausa, Igbo, Pidgin)	None comparable

PECH Recommendation: Faster-Whisper with Large-v3 for server-side transcription, **whisper.cpp with Base/Small** for edge devices. Whisper is the only practical choice for African language coverage.

6. TEXT-TO-SPEECH MODELS

For voice assistants, accessibility, customer support bots, and automated notifications.

Model	License	Quality	Languages	VRAM	Download

Piper TTS	MIT	Good	30+ (incl. custom)	CPU	github.com/rhasspy/piper
Kokoro TTS	Apache-2.0	Very Good	English + Asian	1 GB	huggingface.co/hexgrad/Kokoro-82M
Coqui XTTS-v2	MPL-2.0	Excellent	17 languages	2 GB	github.com/coqui-ai/TTS
Edge-TTS	MIT (client)	Good	300+ voices	API	github.com/rany2/edge-tts

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Self-hosted production	Piper TTS	Fastest; runs on CPU; easy to add custom voices	Kokoro TTS
High-quality English	Kokoro TTS	More natural prosody; Apache-2.0	Coqui XTTS-v2 (MPL)
Voice cloning	Coqui XTTS-v2	Only option for custom voice cloning; MPL-2.0 is permissive	—
Quick prototype	Edge-TTS	Free Microsoft API; 300+ voices; no GPU needed	Piper

PECH Recommendation: Piper TTS for all production use (CPU-only, fast, self-hosted). Add **Kokoro TTS** for premium high-quality English output. Consider **Coqui XTTS-v2** if custom brand voice is needed (MPL-2.0 is compatible with commercial use but requires sharing modifications to Coqui's code).

7. VISION & MULTIMODAL MODELS

For image understanding, visual Q&A, document analysis, and solar panel inspection.

Model	Parameters	License	VRAM (Q4)	Capabilities	Download
Qwen2.5-VL-7B	7B	Apache-2.0	5 GB	Image + video understanding	huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct
Qwen2.5-VL-3B	3B	Apache-2.0	2 GB	Image understanding	huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct
Phi-3-Vision	4.2B	MIT	3 GB	Image understanding	huggingface.co/microsoft/Phi-3-vision-128k-instruct
InternVL2-8B	8B	Apache-2.0	5 GB	Image + document understanding	huggingface.co/OpenGVLab/InternVL2-8B

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Solar panel inspection	Qwen2.5-VL-7B	Best overall vision quality; Apache-2.0; video support	InternVL2-8B
Product image analysis	Qwen2.5-VL-3B	Lightweight; good enough for product Q&A; 2 GB	Phi-3-Vision
Document understanding	InternVL2-8B	Strong at reading documents/charts	Qwen2.5-VL-7B
Edge deployment	Qwen2.5-VL-3B	Smallest multimodal; 2 GB Q4	Phi-3-Vision

PECH Recommendation: Qwen2.5-VL-7B for production (solar inspection, product analysis, support), **Qwen2.5-VL-3B** for lightweight/mobile tasks.

8. OCR MODELS

For invoice processing, receipt scanning, document digitization, tax compliance, and ID verification.

Model	License	Languages	GPU Required	Download
PaddleOCR v4	Apache-2.0	80+ languages	Optional (CPU works)	github.com/PaddlePaddle/PaddleOCR
Tesseract OCR 5	Apache-2.0	100+ languages	CPU only	github.com/tesseract-ocr/tesseract
EasyOCR	Apache-2.0	80+ languages	Optional	github.com/JaidedAI/EasyOCR
Surya OCR	GPL-3.0	90+ languages	GPU recommended	*(excluded — GPL license)*

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Invoice/receipt OCR	PaddleOCR v4	Best accuracy; table detection; layout analysis; Apache-2.0	EasyOCR
Simple text extraction	Tesseract 5	Fastest on CPU; most languages; battle-tested	PaddleOCR
Handwriting recognition	PaddleOCR v4	Better handwriting support than Tesseract	EasyOCR
ID card / KYC	PaddleOCR v4	Built-in ID card detection templates	—

PECH Recommendation: PaddleOCR v4 as primary (invoices, receipts, IDs, KYC), **Tesseract 5** as fallback for simple text extraction.

9. EMBEDDING MODELS

For RAG (Retrieval-Augmented Generation), semantic search, similarity matching, and knowledge retrieval.

Model	Dimensions	License	Size	Download
all-MiniLM-L6-v2	384	Apache-2.0	80 MB	huggingface.co/sentence-transformers/all-MiniLM-L6-v2
all-MiniLM-L12-v2	384	Apache-2.0	120 MB	huggingface.co/sentence-transformers/all-MiniLM-L12-v2
e5-small-v2	384	MIT	130 MB	huggingface.co/intfloat/e5-small-v2
e5-base-v2	768	MIT	420 MB	huggingface.co/intfloat/e5-base-v2
e5-large-v2	1024	MIT	1.3 GB	huggingface.co/intfloat/e5-large-v2
BGE-M3	1024	MIT	2.2 GB	huggingface.co/BAAI/bge-m3
Nomic-Embed-Text-v1.5	768	Apache-2.0	550 MB	huggingface.co/nomic-ai/nomic-embed-text-v1.5

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
RAG pipeline (starter)	all-MiniLM-L6-v2	Tiny (80 MB); fast; good enough for Phase 1	e5-small-v2
RAG pipeline (production)	Nomic-Embed-Text-v1.5	Best quality in Apache-2.0 class; 768 dims; Matryoshka support	BGE-M3 (multilingual)
Multilingual	BGE-M3	Best multilingual; supports 100+ languages	e5-large-v2

embeddings		including African	
Product similarity	all-MiniLM-L12-v2	Good balance of speed and quality for product matching	Nomic-Embed

PECH Recommendation: all-MiniLM-L6-v2 for Phase 1, upgrade to **Nomic-Embed-Text-v1.5** for Phase 2 production RAG.

10. COMPUTER VISION / OBJECT DETECTION

For solar panel inspection, warehouse inventory, security, quality control, and logistics.

Model	License	VRAM	Speed	Download
RT-DETR (PaddlePaddle)	Apache-2.0	2-4 GB	Real-time	github.com/PaddlePaddle/PaddleDetection
RT-DETR (Ultralytics)	AGPL-3.0	2-4 GB	Real-time	*(excluded — AGPL wrapper)*
Detectron2	Apache-2.0	4-8 GB	Near-real-time	github.com/facebookresearch/detectron2
DETR (Facebook)	Apache-2.0	2-4 GB	Moderate	huggingface.co/facebook/detr-resnet-50
OpenCV DNN	Apache-2.0	CPU	Fast	github.com/opencv/opencv
YOLOv5	GPL-3.0	2-4 GB	Real-time	*(excluded — GPL license)*
YOLOv8	AGPL-3.0	2-4 GB	Real-time	*(excluded — AGPL license)*

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Solar panel defect detection	RT-DETR (PaddlePaddle)	Real-time; Apache-2.0; better accuracy than YOLO on small objects	Detectron2
Warehouse inventory	RT-DETR (PaddlePaddle)	Fast counting and detection	DETR
General object detection	Detectron2	Most flexible; strong ecosystem; instance segmentation	RT-DETR
Lightweight / edge	OpenCV DNN	CPU-only; deploy anywhere	—

PECH Recommendation: RT-DETR via PaddleDetection for all computer vision tasks. It replaces YOLOv8 with no license issues and comparable performance. Use **Detectron2** for advanced segmentation tasks.

> **IMPORTANT:** Use the **PaddlePaddle RT-DETR** implementation, NOT the Ultralytics wrapper (which is AGPL).

11. TRADITIONAL ML MODELS

For tabular data tasks: fraud detection, pricing, demand forecasting, credit scoring, anomaly detection.

Model/Library	License	Use Case	Download
XGBoost	Apache-2.0	Classification, regression, ranking	github.com/dmlc/xgboost
LightGBM	MIT	Fast gradient boosting; large datasets	github.com/microsoft/LightGBM

CatBoost	Apache-2.0	Categorical features; less preprocessing	github.com/catboost/catboost
scikit-learn	BSD-3	All-purpose ML toolkit	github.com/scikit-learn/scikit-learn
Prophet	MIT	Time-series forecasting	github.com/facebook/prophet
statsmodels	BSD-3	Statistical modeling	github.com/statsmodels/statsmodels
Isolation Forest	BSD-3 (scikit-learn)	Anomaly/fraud detection	Part of scikit-learn

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Fraud detection	XGBoost + Isolation Forest	XGBoost for classification, IF for anomaly detection	LightGBM
Dynamic pricing	LightGBM	Fastest training; handles large catalogs	XGBoost
Credit scoring	XGBoost	Most interpretable GBM; regulatory compliance	CatBoost
Demand forecasting	Prophet	Built for business time series; handles seasonality	statsmodels ARIMA
Solar yield prediction	LightGBM	Fast retraining with weather data	XGBoost

PECH Recommendation: Use **XGBoost** as the primary ML workhorse, **LightGBM** for speed-critical tasks, **Prophet** for all time-series forecasting.

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12. RECOMMENDATION MODELS

For product recommendations, content suggestions, installer matching, and personalization.

Model/Library	License	Type	Download
LightFM	Apache-2.0	Hybrid (collaborative + content)	github.com/lyst/lightfm
RecBole	MIT	Framework (30+ algorithms)	github.com/RUCAIBox/RecBole
Implicit	MIT	Collaborative filtering (ALS)	github.com/benfred/implicit
Surprise	BSD-3	Classic recommendation algorithms	github.com/NicolasHug/Surprise

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Product recommendations	LightFM	Handles cold-start (new products); uses item metadata	RecBole
"Customers also bought"	Implicit	Fast ALS; GPU-accelerated	LightFM
Installer matching	LightFM	Uses both installer features + customer preferences	—
Research / experimentation	RecBole	30+ algorithms to test	Surprise

PECH Recommendation: LightFM for all production recommendations (marketplace, solar installers, logistics partners). Use **RecBole** for experimentation when optimizing recommendation quality.

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13. IMAGE PROCESSING / GENERATION

For marketplace product images: background removal, enhancement, AI-generated descriptions from images.

Tool	License	Function	Download
rembg	MIT	Background removal	github.com/danielgatis/rembg
U²-Net	Apache-2.0	Salient object detection (powers rembg)	github.com/xuebinqin/U-2-Net
OpenCV	Apache-2.0	Image processing, filters, resize	github.com/opencv/opencv
Pillow	HPND (permissive)	Python image library	github.com/python-pillow/Pillow
Real-ESRGAN	BSD-3	Image upscaling / super-resolution	github.com/xinntao/Real-ESRGAN

PECH Use Cases

Feature	Tool	Where Used
Product background removal	rembg + U²-Net	Marketplace, real estate
Image enhancement	Real-ESRGAN + OpenCV	Product photos, listings
Thumbnail generation	OpenCV + Pillow	All platforms
Image-to-description	Qwen2.5-VL-7B	Marketplace AI features

PECH Recommendation: rembg for background removal (Canva-like feature in marketplace), **Real-ESRGAN** for upscaling low-quality product images, **Qwen2.5-VL** for AI-generated product descriptions from images.

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14. TRANSLATION MODELS

For multilingual support across Nigerian and African languages.

Model	License	Languages	Size	Download
Argos Translate	MIT	30+ language pairs	~100 MB/pair	github.com/argosopentech/argos-translate
NLLB-200	CC-BY-NC-4.0	200 languages	600M-3.3B	*(excluded — non-commercial)*
OPUS-MT	MIT/CC-BY-4.0	1000+ pairs	~300 MB/pair	github.com/Helsinki-NLP/Opus-MT
Whisper (translation mode)	MIT	100+ → English	1.5B	Same as STT above

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
General translation	Argos Translate	MIT; offline; easy to deploy	OPUS-MT
African languages	Whisper (translate mode)	Best coverage of Nigerian languages → English	—
UI/interface translation	Argos Translate	Fast; lightweight; installable per-language	—

PECH Recommendation: Argos Translate for general text translation, **Whisper translate mode** for speech translation from Nigerian languages to English.

> **Note:** African language NLP is still developing. For Yoruba, Hausa, Igbo — Whisper has basic support.

> For production-quality translation in these languages, consider contributing to or fine-tuning on local datasets (Phase 3).

15. NLP MODELS (Classification, NER, Sentiment)

For text classification, named entity recognition, sentiment analysis, and intent detection.

Model	License	Parameters	Use Case	Download
DistilBERT	Apache-2.0	66M	Fast classification, sentiment	huggingface.co/distilbert-base-uncased
BERT-base	Apache-2.0	110M	NER, classification	huggingface.co/google-bert/bert-base-uncased
DeBERTa-v3-small	MIT	44M	Better classification quality	huggingface.co/microsoft/deberta-v3-small
Jais-13B	Apache-2.0	13B	Arabic + English bilingual	huggingface.co/inception-mbzuai/jais-13b

Comparison & Recommendation

Use Case	Best Choice	Why	Runner-Up
Sentiment analysis	DistilBERT	Fastest; 66M params; CPU-friendly	DeBERTa-v3-small
Named entity recognition	BERT-base	Best NER ecosystem; many pre-trained NER heads	DeBERTa-v3-small
Intent classification	DeBERTa-v3-small	Higher accuracy than DistilBERT; still small	DistilBERT
Support ticket routing	DistilBERT	Fast inference; fine-tune on PECH support categories	—

PECH Recommendation: DistilBERT for all lightweight NLP tasks (sentiment, routing, classification). Fine-tune on PECH-specific data in Phase 2. Use **DeBERTa-v3-small** if DistilBERT accuracy is insufficient.

16. MODEL DEPLOYMENT SUMMARY

Phase 1 — Starter Hardware (1× RTX 4090, 24 GB VRAM)

Deploy via **Ollama** (simplest management):

Slot	Model	VRAM	Purpose
Primary LLM	Qwen2.5-7B (Q4)	4.5 GB	General AI assistant
Code	Qwen2.5-Coder-7B (Q4)	4.5 GB	Developer platform
Vision	Qwen2.5-VL-3B (Q4)	2 GB	Image understanding
Embeddings	all-MiniLM-L6-v2	0.1 GB	RAG pipeline
STT	Faster-Whisper (Small)	1 GB	Voice input
TTS	Piper TTS	CPU	Voice output
OCR	PaddleOCR v4	CPU	Invoice/receipt processing

Detection	RT-DETR-L	2 GB	Solar panel inspection
NLP	DistilBERT	CPU	Sentiment/classification
ML	XGBoost + Prophet	CPU	Pricing, forecasting
Total GPU		**~14 GB**	Fits in 24 GB with headroom

Phase 2 — Production (2× RTX 4090 or cloud)

Migrate to **vLLM** (higher throughput, batching):

Slot	Model	VRAM	Purpose
Primary LLM	Qwen2.5-14B (Q4)	8 GB	Better quality assistant
Code	Qwen2.5-Coder-14B (Q4)	8 GB	Full coding assistant
Vision	Qwen2.5-VL-7B (Q4)	5 GB	Production vision
Embeddings	Nomic-Embed-Text-v1.5	0.5 GB	Better RAG quality
Reasoning	Qwen2.5-32B (Q4)	18 GB	Complex analysis (GPU 2)
STT	Faster-Whisper (Large-v3)	4 GB	Best accuracy
All others	Same as Phase 1		

Phase 3 — Scale (Cloud GPU cluster)

- Fine-tuned PECH models on business data
- Qwen2.5-72B (Q4) for premium tier
- Multiple model replicas for concurrency
- A/B testing between model versions

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17. VRAM QUICK-REFERENCE

All measurements assume inference only (not training). Q4 = 4-bit quantization via GGUF/AWQ.

Model	FP16	Q8	Q4	Minimum GPU
Qwen2.5-0.5B	1 GB	600 MB	350 MB	Any GPU / CPU
Qwen2.5-1.5B	3 GB	1.7 GB	1 GB	GTX 1060+
Qwen2.5-7B	14 GB	8 GB	4.5 GB	RTX 3060 12GB
Qwen2.5-14B	28 GB	16 GB	8 GB	RTX 4090 24GB
Qwen2.5-32B	64 GB	36 GB	18 GB	2× RTX 4090
Qwen2.5-72B	144 GB	80 GB	40 GB	A100 80GB or 2× A6000
Mixtral-8x7B	93 GB	52 GB	26 GB	2× RTX 4090
Mistral-7B	14 GB	8 GB	4.5 GB	RTX 3060 12GB
Phi-4 (14B)	28 GB	16 GB	8 GB	RTX 4090 24GB
Phi-3-Mini (3.8B)	7.6 GB	4.3 GB	2.3 GB	RTX 3060 12GB
SmolLM2-1.7B	3.4 GB	1.9 GB	1 GB	Any GPU
Whisper Large-v3	4 GB	2.5 GB	—	RTX 3060+
DistilBERT	260 MB	—	—	CPU
all-MiniLM-L6-v2	80 MB	—	—	CPU

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